

1 Assignment 0

1.1 Setup

Table 1: Machine and Software Configuration

Component	Specification
CPU Model	Intel(R) Core(TM) i5-10310U CPU @ 1.70GHz
Physical Cores	4
Logical Processors / Threads	8
RAM Amount	23.61 GB
SSD Amount	476.94 GB
OS Version	10.0.26200 N/A Build 26200
PostgreSQL Engine Version	PostgreSQL 18.6

1.2 Database

Name: Join Order Benchmark (2013 Snapshot)

This database is based on the 2013 snapshot. The database contains 21 tables.

Table 2: JOB Database Tables, Primary Keys and Cardinalities

Table Name	Primary Key	Cardinality
aka_name	id	901,343
aka_title	id	361,472
cast_info	id	36,244,344
char_name	id	3,140,339
comp_cast_type	id	4
company_name	id	234,997
company_type	id	4
complete_cast	id	135,086
info_type	id	113
keyword	id	134,170
kind_type	id	7
link_type	id	18
movie_companies	id	2,609,129
movie_info	id	14,835,720
movie_info_idx	id	1,380,035
movie_keyword	id	4,523,930
movie_link	id	29,997
name	id	4,167,491
person_info	id	2,963,664
role_type	id	12
title	id	2,528,312

1 Assignment 1

PostgreSQL maintains column statistics after the `ANALYZE` command. For this question, the histograms for the `id` and `title` columns of the `title` table were examined.

1.1 Question 1

1.1.1 1. Corresponding SQL

The statistics were first generated using:

```
ANALYZE title;
```

The histogram information was then obtained from the PostgreSQL `pg_stats` view using:

```
SELECT
    attname,
    n_distinct,
    most_common_vals,
    most_common_freqs,
    histogram_bounds
FROM pg_stats
WHERE tablename = 'title'
    AND attname IN ('id', 'title');
```

The `histogram_bounds` column contains the values used as histogram boundaries. PostgreSQL exposes these statistics through `pg_stats`; the underlying statistics are produced by `ANALYZE`.

1.1.2 2. Type of Histogram

The histogram maintained by PostgreSQL for these columns is an **equi-depth (equal-frequency) histogram**.

The histogram boundaries divide the values into ranges containing approximately the same number of tuples. Therefore, the widths of the value ranges need not be equal.

For example, if a histogram contains B buckets, each bucket approximately represents

$$\frac{1}{B}$$

of the tuples, subject to the effect of most-common values and the sampling used by `ANALYZE`.

1.1.3 3. Number of Buckets

The number of histogram buckets is one less than the number of entries in the `histogram_bounds` array.

For my PostgreSQL statistics:

- `id`: 100
- `title`: 100

The number can be obtained from PostgreSQL using:

```
SELECT
    attname,
    cardinality(histogram_bounds) - 1 AS number_of_buckets
FROM pg_stats
WHERE tablename = 'title'
AND attname IN ('id', 'title');
```

1.1.4 4. How the Textual Attribute `title` is Put into the Histogram

The `title` attribute is a textual value, so it cannot be treated as a numeric value directly.

PostgreSQL first orders the text values according to the column's comparison rules and stores selected ordered values as histogram boundaries. These boundaries divide the ordered text domain into approximately equal-frequency ranges.

For a range predicate on a text column, PostgreSQL converts the text value and the surrounding histogram boundaries to a common scalar comparison scale. It then performs interpolation inside the corresponding histogram bucket.

For strings, PostgreSQL uses the byte representation of the strings as a fractional scalar representation. Common prefixes are removed before the interpolation calculation so that the comparison can focus on the portion of the strings that differs.

Thus, a text histogram does not divide strings according to a numerical magnitude. Instead, it uses the ordering of the strings and a scalar representation suitable for interpolation.

1.1.5 5. Histogram Bucket Boundaries

The bucket boundaries obtained from `pg_stats` are listed below.

`id histogram`

```
{
    58, 24549, 51112, 77316, 100354, 127446, 152219, 175234,
    199351, 226185, 252917, 274611, 298665, 323475, 348752,
    373432, 399278, 422974, 447102, 472333, 496158, 520173,
```

```

546680, 576310, 602304, 628338, 653665, 681361,
707128, 729809, 756258, 782201, 807417, 835118, 863206,
887242, 911779, 936011, 960911, 985388, 1011785, 1035950,
1058604, 1086417, 1112274, 1136653,
1160531, 1186749, 1211107, 1235379, 1259253, 1281711,
1305714, 1330264, 1356923, 1381561, 1405457, 1431660,
1455036, 1481513, 1505512, 1534241, 1559580, 1587278,
1611445, 1637249, 1665180, 1690643, 1715930, 1740280,
1766040, 1789855, 1812673,
1837244, 1864195, 1889430, 1914964, 1939219, 1964987,
1991435, 2015811, 2042317, 2069297, 2095340, 2121752,
2148408, 2173353, 2197551, 2222369, 2247242, 2274114,
2303748, 2328246, 2354510, 2379115, 2403508, 2430141,
2456620, 2480455, 2505196, 2528302
}

```

For example, the boundaries should be interpreted as

$$[b_0, b_1, b_2, \dots, b_B]$$

where bucket i covers the ordered range

$$[b_{i-1}, b_i].$$

title histogram

```

{
  "'A' de abuelo",
  (#1.145), (#1.2686), (#1.4692), (#1.7554),
  (#13.10), (#2.45), (#30.6), (#50.84), (#9.124),
  (1974-03-15), (1989-07-11), (1996-06-14),
  (1999-05-31), (2001-03-21), (2002-12-09),
  (2004-07-16), (2005-11-04), (2007-02-22),
  (2008-06-11), (2010-03-01), (2011-05-21),
  (2012-06-14), (2013-05-16),

  "A Better Tomorrow",
  "A Sculptor's Landscape",
  "Aflevering 421",
  "ALS Video #39: Jamaica I Shoot - Part 2",
  "Anonymous Letters",
  "Auf dem Teppich",
  "Barry Bracelet Boffo",
  "Beware of the Oglethorpe",
  "Bob's Videos 103: Soft & Sexy",
  "Budva na pjenu od mora",
  "Carlos Jr.",

```

"Child Support Assurance",
 Collecting/Chores,
 "Crashus Maximus",
 "Dark Age of Light: Episode 01",
 "Death Tide",
 "Descubra México: Quintana Roo",
 Directions,
 "Dream No Evil",
 "El abrazo de los peces",
 "Enter Another Dragon",
 Faces,
 Finale,
 "Frances and Frankie",
 "Gay Cuba",
 "Going Places",
 "Haha no tomoshibi chiisaki mono wo terashite:
 Ishii fudeko chiteki shōgai ji kyōiku no michi",
 Hennè,
 Homespun,
 "I Have a Stomachache.",
 "In the Night Garden...",
 "J2: Rise of the Dragons",
 "Josh and Cindy's Wedding Trip",
 "Ke arteko egunak",
 "Kuan yin",
 "La maestrina",
 "Le barbier de Séville",
 "Lexi Gets Stiffed",
 "Los Elegidos",
 "Magic Hat",
 "Massacres: New Poetry for Remembering",
 "Miguel Returns to Ella",
 "Motown Maurice Hustle & Shuffle",
 "Nakuna inunaki trio",
 "Nis, Radojkovic",
 Ochtendverkeer,
 "Operation Makeshift",
 "Parfum troublant",
 "Pierde a los dos",
 "Power Rangers Time Force: The End of Time",
 "Qué lolas",
 Rejection,
 "Roinilan talossa",
 "San Mateo, CA",
 "Sehnsucht nach Liebe",
 "Show Forty Five",

```

Socialdemokratiet,
Stacja,
"Summer Tour: MITP Performance 1",
"Tango of Togetherness",
"The American Soldier in Love and War: Number 2",
"The Case of the Golden Fraud",
"The Emperor's Nightingale",
"The Ground We Stand On",
"The Legend of Robin Hood",
"The New Neighbors",
"The Rent Racket: Part 2",
"The Tale of a Feather",
"Them Was the Happy Days!",
"Tödliches Viereck",
"True Cost of Coal",
"Una nevicata a Milano",
"Vendetta I",
"Warsaw Rebuilds",
"Who Am I? Asian Pacific and Proud",
"Xtreme Fakeover",
"Zywot czlowieka rozbrojonego"
}

```

The values are ordered lexicographically according to PostgreSQL's comparison rules for the `title` data type.

1.1.6 6. Histogram Bar Plots

The histograms were visualized using the bucket boundaries on the X-axis and the corresponding frequency on the Y-axis, as required in the assignment.

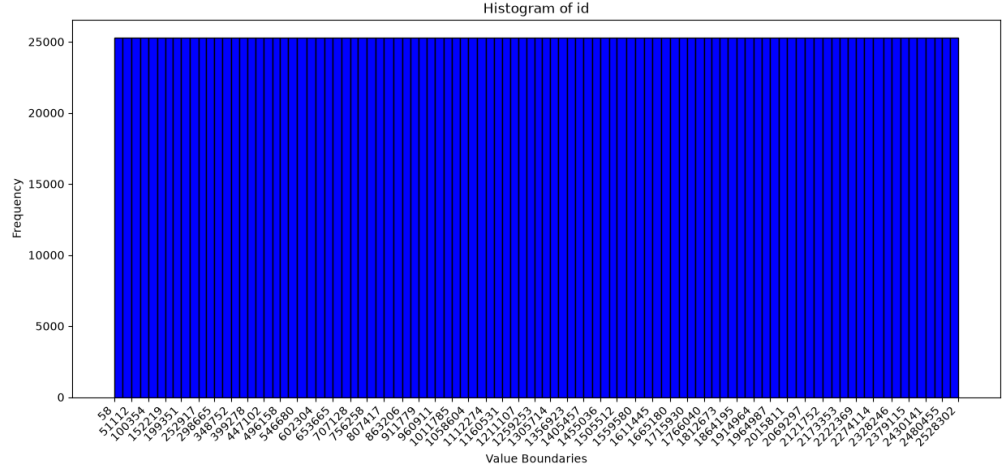


Figure 1: PostgreSQL histogram for `title.id`.

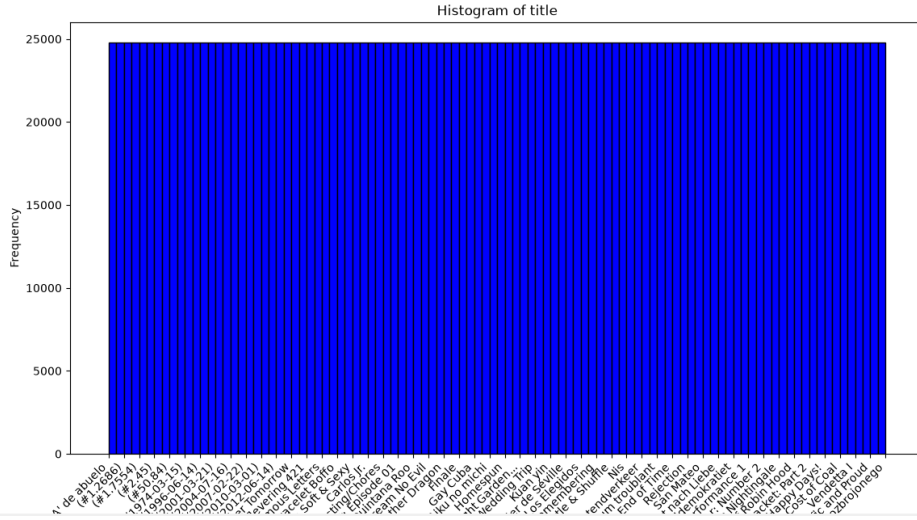


Figure 2: PostgreSQL histogram for `title.title`.

1.1.7 7. Maximum Selectivity Error

For a histogram bucket, PostgreSQL assumes an approximately uniform distribution of values within that bucket.

For a query value q falling inside a bucket with lower and upper boundaries L and U , the estimated fraction inside the bucket is

$$f = \frac{q - L}{U - L}.$$

If the bucket contains an equal fraction of the tuples, the estimated selectivity is obtained by adding all complete preceding buckets and the interpolated fraction of the current bucket.

For a histogram with B buckets, if q lies in bucket i , the estimated selectivity can be represented as

$$\hat{S}(q) = \frac{i - 1 + f}{B}.$$

The actual selectivity is calculated directly from the table:

$$S(q) = \frac{|\{t \mid t.A < q\}|}{|R|}.$$

Therefore, the absolute selectivity error is

$$E(q) = \left| \hat{S}(q) - S(q) \right|.$$

The maximum selectivity error is

$$E_{\max} = \max_q E(q).$$

For my results:

- **id:**

$$E_{\max} = \mathbf{0.01}$$

- **title:**

$$E_{\max} = \mathbf{0.01}$$

The error was calculated by comparing the histogram-based estimate against the actual selectivity obtained from the complete **title** table.

1.1.8 8. Selectivity Estimation for a Text Range Query

For a range query such as

WHERE title < 'Race'

PostgreSQL first locates the histogram bucket containing the value **'Race'**.

It then converts the query value and the two surrounding text boundaries to a common scalar scale. If the query value is between the two boundaries, PostgreSQL assumes an approximately uniform distribution inside that bucket and linearly interpolates the fraction of the bucket covered by the predicate.

If the query value lies before the first boundary or after the last boundary, the corresponding edge case is handled directly.

Thus, the selectivity is approximately

complete preceding buckets + fraction of the current bucket.

This allows PostgreSQL to estimate range predicates even though `title` is a textual attribute.

1.1.9 9. Estimated and Actual Selectivities

Query 1: `title < 'Race'`

The query is:

```
SELECT *
FROM title
WHERE title < 'Race';
```

The estimated number of rows can be obtained using:

```
EXPLAIN
SELECT *
FROM title
WHERE title < 'Race';
```

The actual number of rows is obtained using:

```
SELECT COUNT(*)
FROM title
WHERE title < 'Race';
```

Let

$$N = 2,528,312$$

be the cardinality of the `title` table.

Then:

$$S_{\text{estimated}} = \frac{\text{estimated rows}}{N}$$

and

$$S_{\text{actual}} = \frac{\text{actual rows}}{N}.$$

My results are:

Quantity	Value
Estimated rows	1,938,726
Actual rows	1,919,892
Estimated selectivity	0.7668
Actual selectivity	0.7594
Absolute error	0.0074

Query 2: `id < 350000`

The query is:

```
SELECT *
FROM title
WHERE id < 350000;
```

The estimated number of rows is obtained using:

```
EXPLAIN
SELECT *
FROM title
WHERE id < 350000;
```

The actual number of rows is obtained using:

```
SELECT COUNT(*)
FROM title
WHERE id < 350000;
```

The selectivities are calculated as

$$S_{\text{estimated}} = \frac{\text{estimated rows}}{2,528,312}$$

and

$$S_{\text{actual}} = \frac{\text{actual rows}}{2,528,312}.$$

My results are:

Quantity	Value
Estimated rows	355245
Actual rows	349999
Estimated selectivity	0.1405
Actual selectivity	0.1384
Absolute error	0.0021

2 Q2. Optimal Serial Histogram

For this experiment, an optimal serial histogram was constructed using 20 buckets. The histogram minimizes the objective

$$\sum_i n_i V_i,$$

where n_i is the number of distinct values in bucket i and V_i is the variance of the frequencies within that bucket.

The histograms were constructed for the `id` and `title` attributes of the `title` table.

3 Q2.1 – Histograms for 3000 Samples

The requested sample size was 3000 rows. Due to the use of `TABLESAMPLE`, the number of distinct sampled rows.

The resulting optimal serial histograms contain 20 buckets.

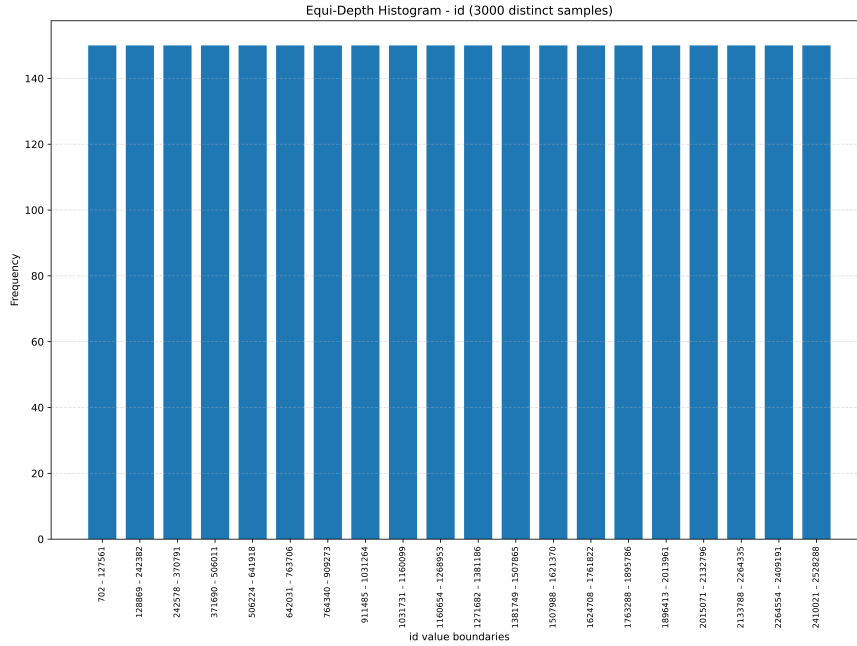


Figure 1: Optimal 20-bucket serial histogram for `id` using the 3000-row requested sample.

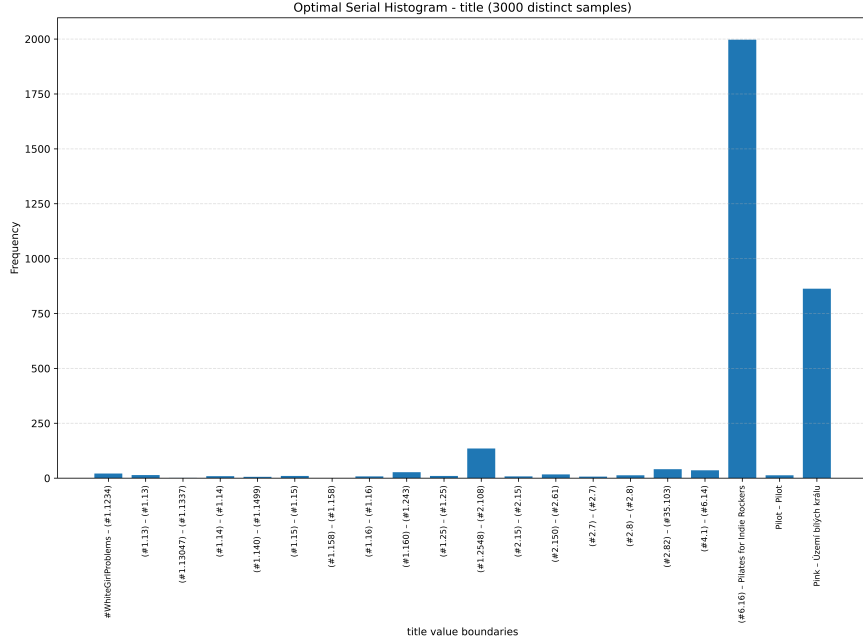


Figure 2: Optimal 20-bucket serial histogram for `title` using the 3000-row requested sample.

For the `id` attribute, all sampled values occur with frequency 1. Consequently, every bucket has mean frequency 1 and zero variance.

For the `title` attribute, most sampled titles occur once, while a smaller number of titles have higher frequencies. The optimization therefore places many low-frequency values in the early buckets and separates the higher-frequency values into smaller buckets.

4 Q2.2 – Histogram Construction Time

The time measured here represents the time required to construct the optimal serial histogram.

For the requested 3000-row sample

Table 3: Histogram construction time for the 3000-row experiment

Column	Build Time (seconds)
<code>id</code>	0.0010938000013993587
<code>title</code>	14.3855853999994

The construction time is higher for `id` in this run, although both attributes contain the same number of sampled rows. The difference is caused by the

different number and distribution of distinct values and frequencies processed during optimization.

5 Q2.3 – Frequency Bucket Boundaries

The frequency bucket boundaries obtained from the optimal 20-bucket serial histograms are shown below.

`id`

Since every sampled `id` occurs exactly once, all 20 frequency buckets have the same frequency boundary:

Table 4: Serial histogram bucket boundaries for `id` using 1000 requested samples.

Bucket	Lower Boundary	Upper Boundary	Number of Values
1	847	140622	50
2	140761	258375	50
3	258615	377423	50
4	378090	484408	50
5	499968	597279	50
6	600699	725692	50
7	731697	871505	50
8	881815	1020829	50
9	1020894	1152758	50
10	1154192	1274886	50
11	1276675	1382541	50
12	1383813	1486390	50
13	1486727	1615032	50
14	1615812	1770041	50
15	1770183	1907517	50
16	1911664	2039649	50
17	2046649	2167311	50
18	2172668	2298942	50
19	2299223	2400490	50
20	2403173	2527908	50

Table 5: Serial histogram bucket boundaries for `id` using 3000 requested samples.

Bucket	Lower Boundary	Upper Boundary	Number of Values
1	702	127561	150
2	128869	242382	150
3	242578	370791	150
4	371690	506011	150
5	506224	641918	150
6	642031	763706	150
7	764340	909273	150
8	911485	1031264	150
9	1031731	1160099	150
10	1160654	1268953	150
11	1271682	1381186	150
12	1381749	1507865	150
13	1507988	1621370	150
14	1624708	1761822	150
15	1763288	1895786	150
16	1896413	2013961	150
17	2015071	2132796	150
18	2133788	2264335	150
19	2264554	2409191	150
20	2410021	2528288	150

Table 6: Serial histogram bucket boundaries for `id` using 5000 requested samples.

Bucket	Lower Boundary	Upper Boundary	Number of Values
1	131	124118	250
2	124198	255071	250
3	256015	383393	250
4	384002	514336	250
5	514451	635485	250
6	635704	769108	250
7	769553	904333	250
8	904394	1027133	250
9	1028395	1151816	250
10	1152319	1293390	250
11	1294188	1419088	250
12	1419183	1545279	250
13	1545904	1670242	250
14	1670446	1786731	250
15	1787355	1919436	250
16	1919878	2046085	250
17	2046179	2160856	250
18	2160898	2283435	250
19	2283506	2410390	250
20	2410625	2527725	250

Table 7: Serial histogram bucket boundaries for `title` using 1000 requested samples.

Bucket	Lower Boundary	Upper Boundary	Number of Values
1	(#1.1)	(#1.1)	22
2	(#1.10627)	(#1.109)	3
3	(#1.11)	(#1.11)	8
4	(#1.111)	(#1.11887)	6
5	(#1.12)	(#1.12)	4
6	(#1.124)	(#1.208)	11
7	(#1.22)	(#1.23)	7
8	(#1.24)	(#1.5258)	27
9	(#1.6)	(#1.6)	15
10	(#1.60)	(#19.110)	20
11	(#2.1)	(#2.1)	5
12	(#2.14)	(#2.18)	8
13	(#2.2)	(#2.2)	9
14	(#2.32)	(#2.42)	3
15	(#2.5)	(#2.7)	10
16	(#26.37)	(#3.27)	11
17	(#3.3)	(#3.4)	7
18	(#3.43)	(#3.43)	1
19	(#3.7)	(#3.8)	7
20	(#31.18)	Ôgon no meiyaku	918

Table 8: Serial histogram bucket boundaries for `title` using 3000 requested samples.

Bucket	Lower Boundary	Upper Boundary	Number of Values
1	#WhiteGirlProblems	(#1.1234)	21
2	(#1.13)	(#1.13)	14
3	(#1.13047)	(#1.1337)	2
4	(#1.14)	(#1.14)	9
5	(#1.140)	(#1.1499)	6
6	(#1.15)	(#1.15)	10
7	(#1.158)	(#1.158)	1
8	(#1.16)	(#1.16)	8
9	(#1.160)	(#1.243)	27
10	(#1.25)	(#1.25)	10
11	(#1.2548)	(#2.108)	135
12	(#2.15)	(#2.15)	8
13	(#2.150)	(#2.61)	17
14	(#2.7)	(#2.7)	7
15	(#2.8)	(#2.8)	13
16	(#2.82)	(#35.103)	41
17	(#4.1)	(#6.14)	36
18	(#6.16)	Pilates for Indie Rockers	1997
19	Pilot	Pilot	13
20	Pink	Území bílých králu	863

Table 9: Serial histogram bucket boundaries for `title` using 5000 requested samples.

Bucket	Lower Boundary	Upper Boundary	Number of Values
1	'31 Truck Part 3	(#1.10774)	24
2	(#1.11)	(#1.11)	34
3	(#1.11126)	(#1.1278)	9
4	(#1.13)	(#1.13)	19
5	(#1.1307)	(#1.15662)	20
6	(#1.16)	(#1.16)	17
7	(#1.1600)	(#1.1686)	5
8	(#1.17)	(#1.17)	25
9	(#1.174)	(#1.198)	20
10	(#1.2)	(#1.2)	108
11	(#1.202)	(#1.286)	39
12	(#1.29)	(#1.29)	13
13	(#1.295)	(#1.2980)	4
14	(#1.3)	(#1.3)	93
15	(#1.311)	(#2.45)	234
16	(#2.7)	(#2.7)	17
17	(#2.77)	(#2.9)	28
18	(#20.122)	Pikavieraita	3482
19	Pilot	Pilot	27
20	Pimple Goes A-Busking	Útvesztöben	1511

Thus, the optimal partition groups the large number of frequency-1 values together and allocates progressively smaller buckets to the less frequent but higher-frequency levels.

6 Q2.4 – Maximum Selectivity Error

The selectivity error was evaluated against the actual full-table frequency distribution rather than only against the sample.

For a value v ,

$$S_{\text{actual}}(v) = \frac{f_{\text{full}}(v)}{N_{\text{full}}},$$

and the histogram estimate is

$$S_{\text{estimated}}(v) = \frac{\hat{f}(v)}{N_{\text{full}}},$$

where

$$N_{\text{full}} = 2,528,312.$$

The maximum error is therefore

$$E_{\max} = \max_v |S_{\text{estimated}}(v) - S_{\text{actual}}(v)|.$$

For the `title` histogram, the maximum error was obtained for the title (`#1.1`).

Table 10: Maximum selectivity error for the 3000-row experiment

Measure	id	title
Maximum selectivity error	0.0096363898	0.0409486646

1 Q2.5 – Different Sample Sizes

The same histogram construction procedure was repeated for requested sample sizes of 1000 and 5000.

Because TABLESAMPLE does not necessarily return exactly the requested number of rows, the actual sample sizes were recorded.

Table 1: Histogram construction times for different sample sizes

Requested Size	Build Time (s)	
	id	title
1000	0.0010938	14.3855854
3000	0.0018835	156.4226303
5000	0.0020281	416.7633329

The corresponding serial histograms for the additional sample sizes are shown below.

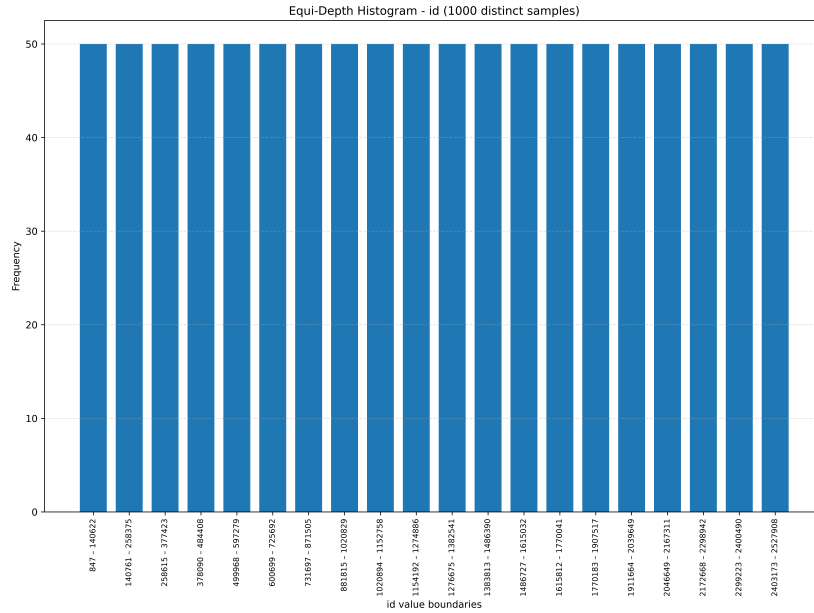


Figure 1: Optimal serial histogram for id using 1000 requested samples.

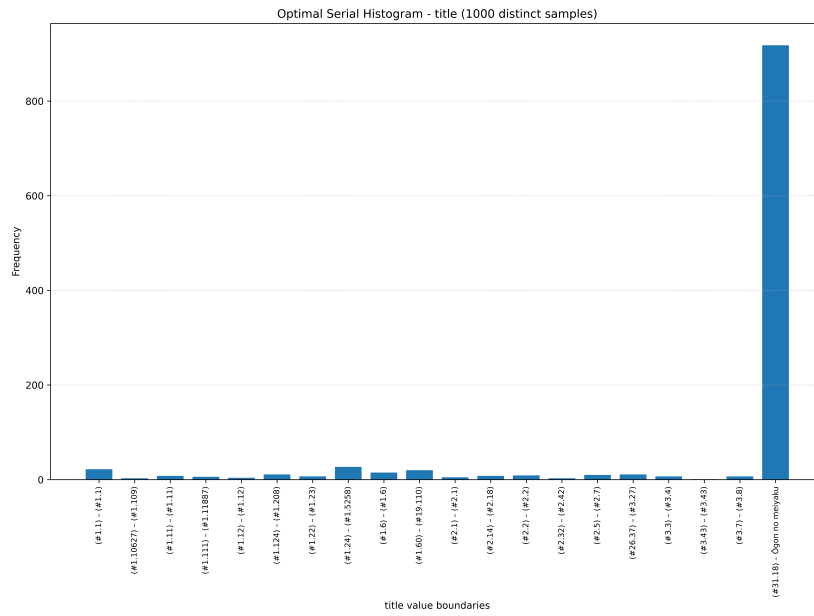


Figure 2: Optimal serial histogram for `title` using 1000 requested samples.

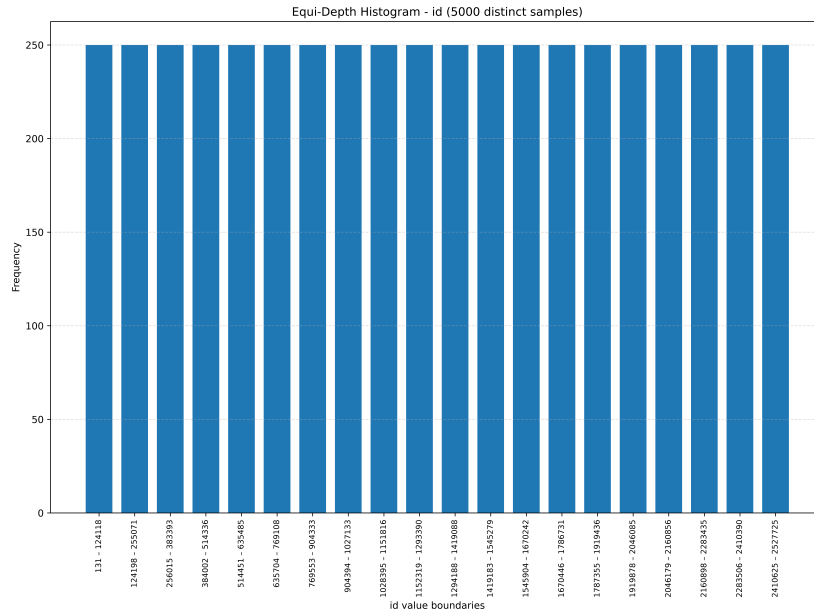


Figure 1: Optimal serial histogram for `id` using 5000 requested samples.

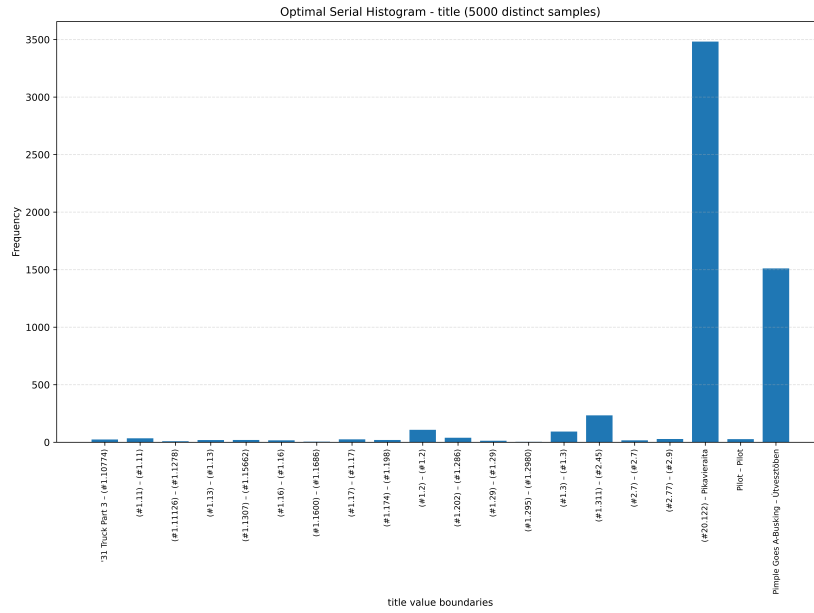


Figure 2: Optimal serial histogram for `title` using 5000 requested samples.

7 Q2.6 – Extrapolation to the Full Table

The full `title` table contains

$$N_{\text{full}} = 2,528,312$$

rows.

A linear-through-origin model was used:

$$T(n) = an,$$

where n is the actual sample size and $T(n)$ is the measured histogram construction time.

The slope is calculated using

$$a = \frac{\sum_i n_i T_i}{\sum_i n_i^2}.$$

the extrapolated times are:

Table 11: Estimated full-table histogram construction time		
Column	Estimated Time (seconds)	Estimated Time (minutes)
<code>id</code>	0.591518	0.009859
<code>title</code>	254228.196	4237.1366

Therefore, the estimated construction time for the complete table is approximately

$$0.000164 \text{ hours}$$

for `id`, and

$$70.618943 \text{ hours}$$

for `title`.

These values are extrapolations based on the measured sample sizes and should not be interpreted as actual execution times for constructing a full-table histogram.

2 Q2.7 – Sample Size vs. Time

The relationship between sample size and histogram construction time is shown in Figure 3.

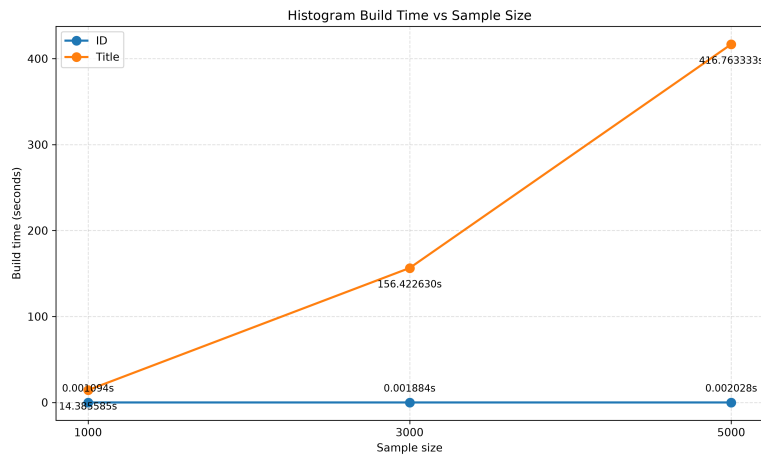


Figure 3: Relationship between sample size and histogram construction time.

The plot shows an increasing trend in construction time as the sample size increases. The increase is particularly noticeable between the 3000- and 5000-sample experiments.

3 Q2.8 – Sample Size vs. Maximum Selectivity Error

The maximum selectivity error is expected to vary with sample size because the histogram is constructed from sampled data. Increasing the sample size generally provides a more representative frequency distribution and can reduce estimation error, although the exact trend depends on the sampled data and histogram partition.

The required relationship is plotted in Figure 4.

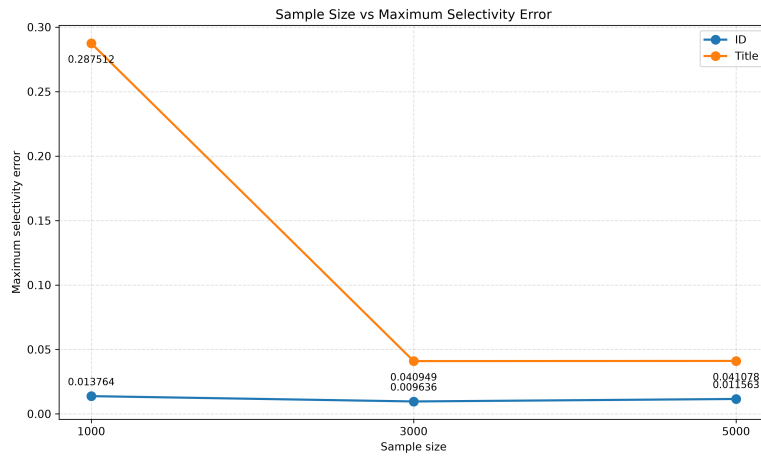


Figure 4: Relationship between sample size and maximum selectivity error.

For the 3000-requested-sample experiment, the corrected maximum selectivity errors were:

Table 1: Verified maximum selectivity error for the 3000-row experiment

Column	Maximum Error
id	0.00963639
title	0.0409487

4 Chats with LLM

<https://chatgpt.com/share/6a95491f-cbe0-83e9-ad01-4a8ed1d2c84e>
<https://chatgpt.com/share/6a9e9344-9404-83e8-bd19-4044372423f7>

5 GitHub Link

<https://github.com/Kulkarnikeyur/ADBMS-Assignment-0-1.git>